

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

— * —



BÁO CÁO BÀI TẬP

BTVN 03 + 04

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B01 : FILE_SERVER

Ảnh minh họa khởi tạo server, client:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./file_server
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c
```

Ảnh minh họa client1 gửi file trong thư mục:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./file_server
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c

acc.txt
Received: OK 24
yuya 123456
admin 123456

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c
```

Ảnh minh họa client2 gửi file không có trong thư mục:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./file_serve
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c

acc.txt
Received: OK 24
yuya 123456
admin 123456

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: OK 13
chat_server.c
client
client.c
http_server
acc.txt
file_server
telnet_server
chat_server
file_server.c
http_server.c
time_server
telnet_server.c
time_server.c

out.txt
Received: ERROR File không tồn tại. Nhập lại:
█
```

Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <sys/types.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <signal.h>
#include <dirent.h>

void handle_client(int client) {
    DIR *d = opendir(".");
    if (!d) {
        close(client);
        return;
    }

    struct dirent *dir;
    char file_list[256][256];
    int count = 0;

    while ((dir = readdir(d)) != NULL) {
        if (dir->d_type == DT_REG) {
```

```

        strcpy(file_list[count], dir->d_name);
        count++;
    }
}
closedir(d);

if (count == 0) {
    char *err = "ERROR No files to download \r\n";
    send(client, err, strlen(err), 0);
    close(client);
    return;
}

char header[64];
snprintf(header, sizeof(header), "OK %d\r\n", count);
send(client, header, strlen(header), 0);

for (int i = 0; i < count; i++) {
    char line[512];
    snprintf(line, sizeof(line), "%s\r\n", file_list[i]);
    send(client, line, strlen(line), 0);
}
send(client, "\r\n", 2, 0);

char buf[256];
while (1) {
    int ret = recv(client, buf, sizeof(buf) - 1, 0);
    if (ret <= 0) break;

    buf[ret] = 0;
    buf[strcspn(buf, "\r\n")] = 0;

    if (strlen(buf) == 0) continue;

    FILE *f = fopen(buf, "rb");
    if (f != NULL) {
        fseek(f, 0, SEEK_END);
        long size = ftell(f);
        fseek(f, 0, SEEK_SET);

        char ok_msg[64];
        snprintf(ok_msg, sizeof(ok_msg), "OK %ld\r\n", size);
        send(client, ok_msg, strlen(ok_msg), 0);

        char file_buf[1024];
        int bytes;
        while ((bytes = fread(file_buf, 1, sizeof(file_buf), f)) > 0) {
            send(client, file_buf, bytes, 0);
        }
        fclose(f);
    }
}

```

```

        break;
    } else {
        char *err_msg = "ERROR File khong ton tai. Nhap lai:\r\n";
        send(client, err_msg, strlen(err_msg), 0);
    }
}
close(client);
}

int main() {
    signal(SIGCHLD, SIG_IGN);

    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    bind(listener, (struct sockaddr *)&addr, sizeof(addr));
    listen(listener, 10);

    while (1) {
        int client = accept(listener, NULL, NULL);
        if (client < 0) continue;

        if (fork() == 0) {
            close(listener);
            handle_client(client);
            exit(0);
        }
        close(client);
    }

    close(listener);
    return 0;
}

```

B02 : GROUP_CHAT_SERVER

Ảnh minh họa khởi tạo server, 1 client kết nối:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./group_chat_server

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Dang cho ghep cap...
█
```

Ảnh minh họa có 2 client kết nối:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./group_chat_server

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Dang cho ghep cap...
Da ghep cap! Bat dau chat:
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Da ghep cap! Bat dau chat:
█
```

Ảnh minh họa 2 client gửi tin nhắn cho nhau:

<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./group_chat_server</pre>	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... Dang cho ghep cap... Da ghep cap! Bat dau chat: hi client 2 hello client 1 █</pre>
	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... Da ghep cap! Bat dau chat: hi client 2 hello client 1 █</pre>

Ảnh minh họa 1 client ngắt kết nối:

<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./group_chat_server</pre>	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... Dang cho ghep cap... Da ghep cap! Bat dau chat: hi client 2 hello client 1 ^C sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$ █</pre>
	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... Da ghep cap! Bat dau chat: hi client 2 hello client 1 [Server da ngat ket noi] sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$ █</pre>

Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <pthread.h>
#include <sys/select.h>

int waiting_client = -1;
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

struct Pair {
    int c1;
    int c2;
};

void *pair_thread(void *arg) {
    struct Pair *p = (struct Pair *)arg;
    int c1 = p->c1;
    int c2 = p->c2;
    free(p);

    char buf[1024];
    fd_set readfds;
    int max_fd = (c1 > c2) ? c1 : c2;

    while (1) {
        FD_ZERO(&readfds);
        FD_SET(c1, &readfds);
        FD_SET(c2, &readfds);

        if (select(max_fd + 1, &readfds, NULL, NULL, NULL) < 0) {
            break;
        }

        if (FD_ISSET(c1, &readfds)) {
            int ret = recv(c1, buf, sizeof(buf), 0);
            if (ret <= 0) break;
            send(c2, buf, ret, 0);
        }

        if (FD_ISSET(c2, &readfds)) {
            int ret = recv(c2, buf, sizeof(buf), 0);
            if (ret <= 0) break;
            send(c1, buf, ret, 0);
        }
    }
}
```

```

    close(c1);
    close(c2);
    return NULL;
}

int main() {
    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    bind(listener, (struct sockaddr *)&addr, sizeof(addr));
    listen(listener, 10);

    while (1) {
        int client = accept(listener, NULL, NULL);
        if (client < 0) continue;

        pthread_mutex_lock(&mutex);
        if (waiting_client == -1) {
            waiting_client = client;
            char *msg = "Dang cho ghep cap...\n";
            send(client, msg, strlen(msg), 0);
            pthread_mutex_unlock(&mutex);
        } else {
            int c1 = waiting_client;
            int c2 = client;
            waiting_client = -1;
            pthread_mutex_unlock(&mutex);

            char *msg = "Da ghep cap! Bat dau chat:\n";
            send(c1, msg, strlen(msg), 0);
            send(c2, msg, strlen(msg), 0);

            struct Pair *p = malloc(sizeof(struct Pair));
            p->c1 = c1;
            p->c2 = c2;

            pthread_t tid;
            pthread_create(&tid, NULL, pair_thread, p);
            pthread_detach(tid);
        }
    }

    close(listener);
    return 0;
}

```

B03 : CHAT_SERVER

Ảnh minh họa client đăng nhập sai:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./chat_server
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
yuya
Received: (client_id: client_name):
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
```

Ảnh minh họa client đăng nhập đúng:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./chat_server
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
yuya
Received: (client_id: client_name):
client_id: yuya
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
```

Ảnh minh họa client gửi tin nhắn cho nhau:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./chat_serve
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
yuya
Received: (client_id: client_name):
client_id: yuya
hello
Received: 2026/05/24 09:58:28PM yuto: hi
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: (client_id: client_name):
client_id: yuto
Received: 2026/05/24 09:58:24PM yuya: hello
hi
█
```

Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <pthread.h>
#include <time.h>

int clients[256];
int num_clients = 0;
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

void *client_thread(void *arg) {
    int client = *(int *)arg;
    free(arg);
    char buf[1024];
    char client_name[64];

    while (1) {
        send(client, "(client_id: client_name): ", 35, 0);
        int ret = recv(client, buf, sizeof(buf) - 1, 0);
```

```

        if (ret <= 0) {
            close(client);
            return NULL;
        }
        buf[ret] = 0;
        buf[strcspn(buf, "\r\n")] = 0;

        if (sscanf(buf, "client_id: %s", client_name) == 1) {
            break;
        }
    }

    pthread_mutex_lock(&mutex);
    clients[num_clients++] = client;
    pthread_mutex_unlock(&mutex);

    while (1) {
        int ret = recv(client, buf, sizeof(buf) - 1, 0);
        if (ret <= 0) break;
        buf[ret] = 0;
        buf[strcspn(buf, "\r\n")] = 0;

        if (strlen(buf) == 0) continue;

        time_t t = time(NULL);
        struct tm *tmp = localtime(&t);
        char time_str[64];
        strftime(time_str, sizeof(time_str), "%Y/%m/%d %I:%M:%S%p", tmp);

        char send_buf[2048];
        snprintf(send_buf, sizeof(send_buf), "%s %s: %s\n", time_str, client_name,
buf);

        pthread_mutex_lock(&mutex);
        for (int i = 0; i < num_clients; i++) {
            if (clients[i] != client) {
                send(clients[i], send_buf, strlen(send_buf), 0);
            }
        }
        pthread_mutex_unlock(&mutex);
    }

    pthread_mutex_lock(&mutex);
    for (int i = 0; i < num_clients; i++) {
        if (clients[i] == client) {
            clients[i] = clients[num_clients - 1];
            num_clients--;
            break;
        }
    }
}

```

```
pthread_mutex_unlock(&mutex);
close(client);
return NULL;
}

int main() {
    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    bind(listener, (struct sockaddr *)&addr, sizeof(addr));
    listen(listener, 10);

    while (1) {
        int client = accept(listener, NULL, NULL);
        if (client < 0) continue;

        pthread_t tid;
        int *arg = malloc(sizeof(int));
        *arg = client;
        pthread_create(&tid, NULL, client_thread, arg);
        pthread_detach(tid);
    }
    close(listener);
    return 0;
}
```

B04 : TELNET_SERVER

Ảnh minh họa client nhập sai mật khẩu:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./telnet_server
ver

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
yuya 123654
Received: ERROR
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
```

Ảnh minh họa client đăng nhập thành công:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./telnet_server
ver

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
yuya 123654
Received: ERROR
yuya 123456
Received: OK
>
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
```

Ảnh minh họa client gửi sai lệnh:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./telnet_server
ver

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
yuya 123654
Received: ERRO
yuya 123456
Received: OK
>
dir
Received: sh: 1: ddir: not found
Received: >
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
```

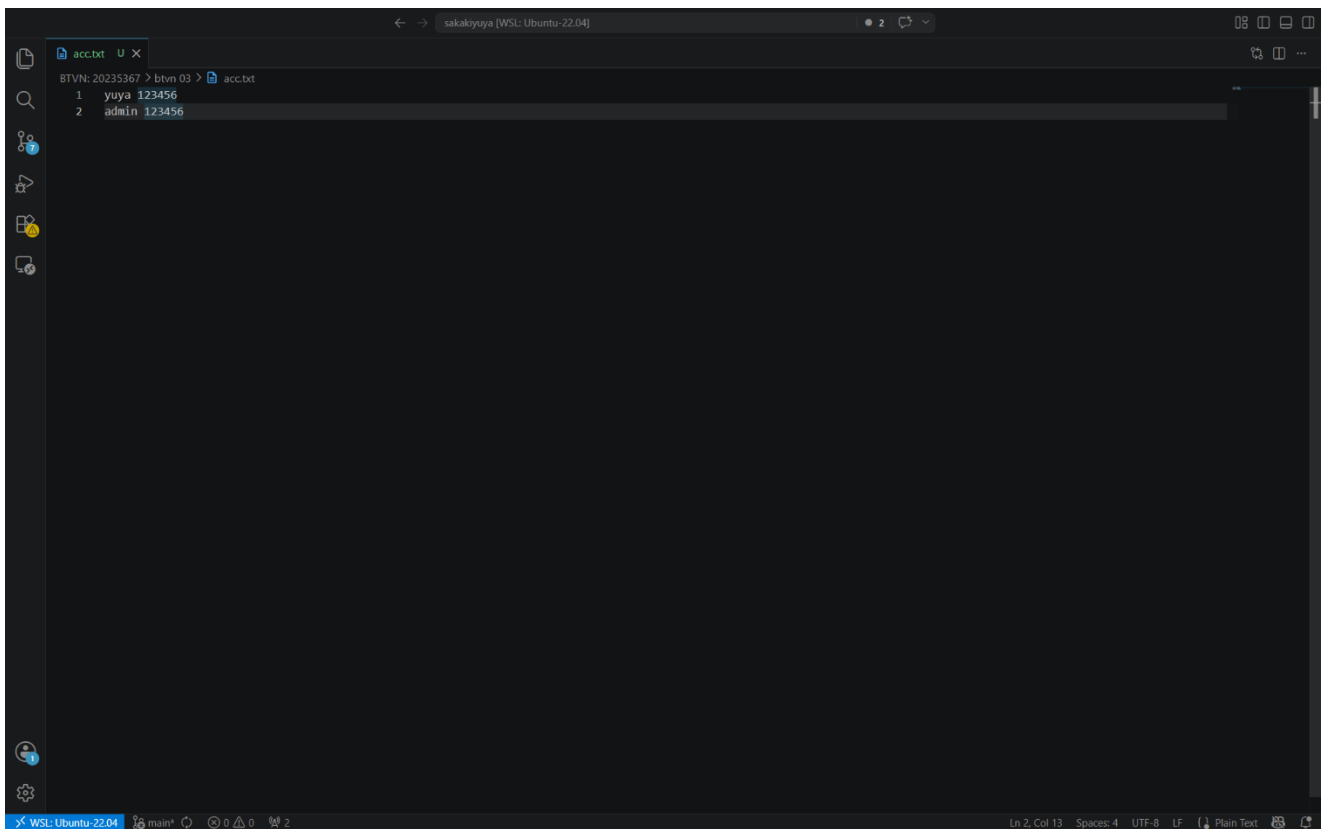
Ảnh minh họa client gửi đúng lệnh:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./telnet_server
ver

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
yuya 123654
Received: ERRO
yuya 123456
Received: OK
>
dir
Received: sh: 1: ddir: not found
Received: >
dir
Received: acc.txt      chat_server.c client.c telnet_server
chat_server client    out_4.txt telnet_server.c
Received: >

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
Received: User and Pass:
admin 123456
Received: OK
>
dir
Received: acc.txt      chat_server.c client.c telnet_server
chat_server client    out_5.txt telnet_server.c
Received: >
█
```

Ảnh file acc.txt:



Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <pthread.h>

char accounts[100][64];
int account_count = 0;

void *client_thread(void *arg) {
    int client = *(int *)arg;
    free(arg);
    char buf[1024] = {0};
    int authenticated = 0;

    send(client, "User and Pass:\n", 14, 0);

    while (1) {
        int ret = recv(client, buf, sizeof(buf) - 1, 0);
        if (ret <= 0) break;
    }
}
```

```

    buf[ret] = '\0';
    buf[strcspn(buf, "\r\n")] = 0;
    if (strlen(buf) == 0) continue;

    if (!authenticated) {
        for (int k = 0; k < account_count; k++) {
            if (strcmp(buf, accounts[k]) == 0) {
                authenticated = 1;
                break;
            }
        }

        if (authenticated) {
            send(client, "OK\n> ", 5, 0);
        } else {
            send(client, "ERROR\n", 4, 0);
        }
    }
    else {
        char command[2048] = {0};
        char filename[64] = {0};

        snprintf(filename, sizeof(filename), "out_%d.txt", client);
        snprintf(command, sizeof(command), "%s > %s 2>&1", buf, filename);
        system(command);

        FILE *f = fopen(filename, "r");
        if (f != NULL) {
            char *line = NULL;
            size_t len = 0;
            while (getline(&line, &len, f) != -1) {
                send(client, line, strlen(line), 0);
            }
            fclose(f);
            if (line) free(line);
            remove(filename);
        }
        send(client, "> ", 2, 0);
    }
}
close(client);
return NULL;
}

int main() {
    FILE *file = fopen("acc.txt", "r");
    if (file) {
        char *line = NULL;
        size_t len = 0;

```

```

    while (getline(&line, &len, file) != -1) {
        line[strcspn(line, "\r\n")] = 0;
        strcpy(accounts[account_count++], line);
    }
    fclose(file);
    if (line) free(line);
}

int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
struct sockaddr_in addr = {0};
addr.sin_family = AF_INET;
addr.sin_addr.s_addr = htonl(INADDR_ANY);
addr.sin_port = htons(8080);

bind(listener, (struct sockaddr *)&addr, sizeof(addr));
listen(listener, 10);

while (1) {
    int client = accept(listener, NULL, NULL);
    if (client < 0) continue;

    pthread_t tid;
    int *arg = malloc(sizeof(int));
    *arg = client;
    pthread_create(&tid, NULL, client_thread, arg);
    pthread_detach(tid);
}
close(listener);
return 0;
}

```

B05 : HTTP_SERVER

Ảnh minh họa khởi tạo server, client:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./http_server
r
New client connected: 4
New client connected: 5
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
```

Ảnh minh họa client1 gửi tin nhắn đến server:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./http_server
r
New client connected: 4
New client connected: 5
hello iam client 1
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
hello iam client 1
Received: HTTP/1.1 200 OK
Content-Type: text/html

Xin chao cac ban
█

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
```

Ảnh minh họa client2 gửi tin nhắn đến server:

<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./http_serve r New client connected: 4 New client connected: 5 hello iam client 1 hi im client 2 █</pre>	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... hello iam client 1 Received: HTTP/1.1 200 OK Content-Type: text/html Xin chao cac ban sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04\$./client Connected to server on port 8080... hi im client 2 Received: HTTP/1.1 200 OK Content-Type: text/html Xin chao cac ban █</pre>
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Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <pthread.h>

int listener;
pthread_mutex_t mutex = PTHREAD_MUTEX_INITIALIZER;

void *worker_thread(void *arg) {
    while (1) {
        pthread_mutex_lock(&mutex);
        int client = accept(listener, NULL, NULL);
        pthread_mutex_unlock(&mutex);

        if (client < 0) continue;

        printf("New client connected: %d\n", client);

        char buf[256];
        int ret = recv(client, buf, sizeof(buf) - 1, 0);
```

```

        if (ret > 0) {
            buf[ret] = 0;
            puts(buf);

            char *msg = "HTTP/1.1 200 OK\r\nContent-Type: text/html\r\n\r\nXin chao
cac ban\n";
            send(client, msg, strlen(msg), 0);
        }

        close(client);
    }
    return NULL;
}

int main() {
    listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    bind(listener, (struct sockaddr *)&addr, sizeof(addr));
    listen(listener, 10);

    pthread_t threads[5];
    for (int i = 0; i < 5; i++) {
        pthread_create(&threads[i], NULL, worker_thread, NULL);
    }

    for (int i = 0; i < 5; i++) {
        pthread_join(threads[i], NULL);
    }

    close(listener);
    return 0;
}

```

B06 : TIME_SERVER

Ảnh minh họa client gửi sai cú pháp:

<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./time_serve r</pre>	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./client Connected to server on port 8080... yyyy/dd/mm Received: Loi: Sai cu phap. Dung: GET_TIME [format] sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./client Connected to server on port 8080...</pre>
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Ảnh minh họa client gửi sai định dạng:

<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./time_serve r</pre>	<pre>sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./client Connected to server on port 8080... yyyy/dd/mm Received: Loi: Sai cu phap. Dung: GET_TIME [format] GET_TIME yyyy/dd/mm Received: Loi: Dinh dang thoi gian khong duoc ho tro. sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btn 03 + 04\$./client Connected to server on port 8080...</pre>
---	---

Ảnh minh họa client gửi đúng cú pháp, định dạng:

```
sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./time_serve
r

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
yyyy/dd/mm
Received: Loi: Sai cu phap. Dung: GET_TIME [format]

GET_TIME yyyy/dd/mm
Received: Loi: Dinh dang thoi gian khong duoc ho tro.

GET_TIME dd/mm/yyyy
Received: 24/05/2026

sakakiyuya@DESKTOP-7E0C4GH:~/BTVN: 20235367/btvn 03 + 04$ ./client
Connected to server on port 8080...
GET_TIME mm/dd/yy
Received: 05/24/26

█
```

Source code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <unistd.h>
#include <pthread.h>
#include <time.h>

void *client_thread(void *arg) {
    int client = *(int *)arg;
    free(arg);

    char buf[256];

    while (1) {
        int ret = recv(client, buf, sizeof(buf) - 1, 0);
        if (ret <= 0) break;

        buf[ret] = '\0';
        buf[strcspn(buf, "\r\n")] = 0;

        if (strlen(buf) == 0) continue;
```

```

char cmd[32] = {0}, format[32] = {0};
int n = sscanf(buf, "%s %s", cmd, format);

if (n < 2 || strcmp(cmd, "GET_TIME") != 0) {
    char *msg = "Loi: Sai cu phap. Dung: GET_TIME [format]\n";
    send(client, msg, strlen(msg), 0);
    continue;
}

char *time_fmt = NULL;
if (strcmp(format, "dd/mm/yyyy") == 0) time_fmt = "%d/%m/%Y";
else if (strcmp(format, "dd/mm/yy") == 0) time_fmt = "%d/%m/%y";
else if (strcmp(format, "mm/dd/yyyy") == 0) time_fmt = "%m/%d/%Y";
else if (strcmp(format, "mm/dd/yy") == 0) time_fmt = "%m/%d/%y";

if (time_fmt != NULL) {
    time_t t = time(NULL);
    struct tm *tmp = localtime(&t);
    char out[64] = {0};

    strftime(out, sizeof(out), time_fmt, tmp);
    strcat(out, "\n");

    send(client, out, strlen(out), 0);
} else {
    char *msg = "Loi: Dinh dang thoi gian khong duoc ho tro.\n";
    send(client, msg, strlen(msg), 0);
}

close(client);
return NULL;
}

int main() {
    int listener = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);

    struct sockaddr_in addr = {0};
    addr.sin_family = AF_INET;
    addr.sin_addr.s_addr = htonl(INADDR_ANY);
    addr.sin_port = htons(8080);

    bind(listener, (struct sockaddr *)&addr, sizeof(addr));
    listen(listener, 10);

    while (1) {
        int client = accept(listener, NULL, NULL);
        if (client < 0) continue;
    }
}

```

```
int *arg = malloc(sizeof(int));
*arg = client;

pthread_t tid;
pthread_create(&tid, NULL, client_thread, arg);
pthread_detach(tid);
}

close(listener);
return 0;
}
```